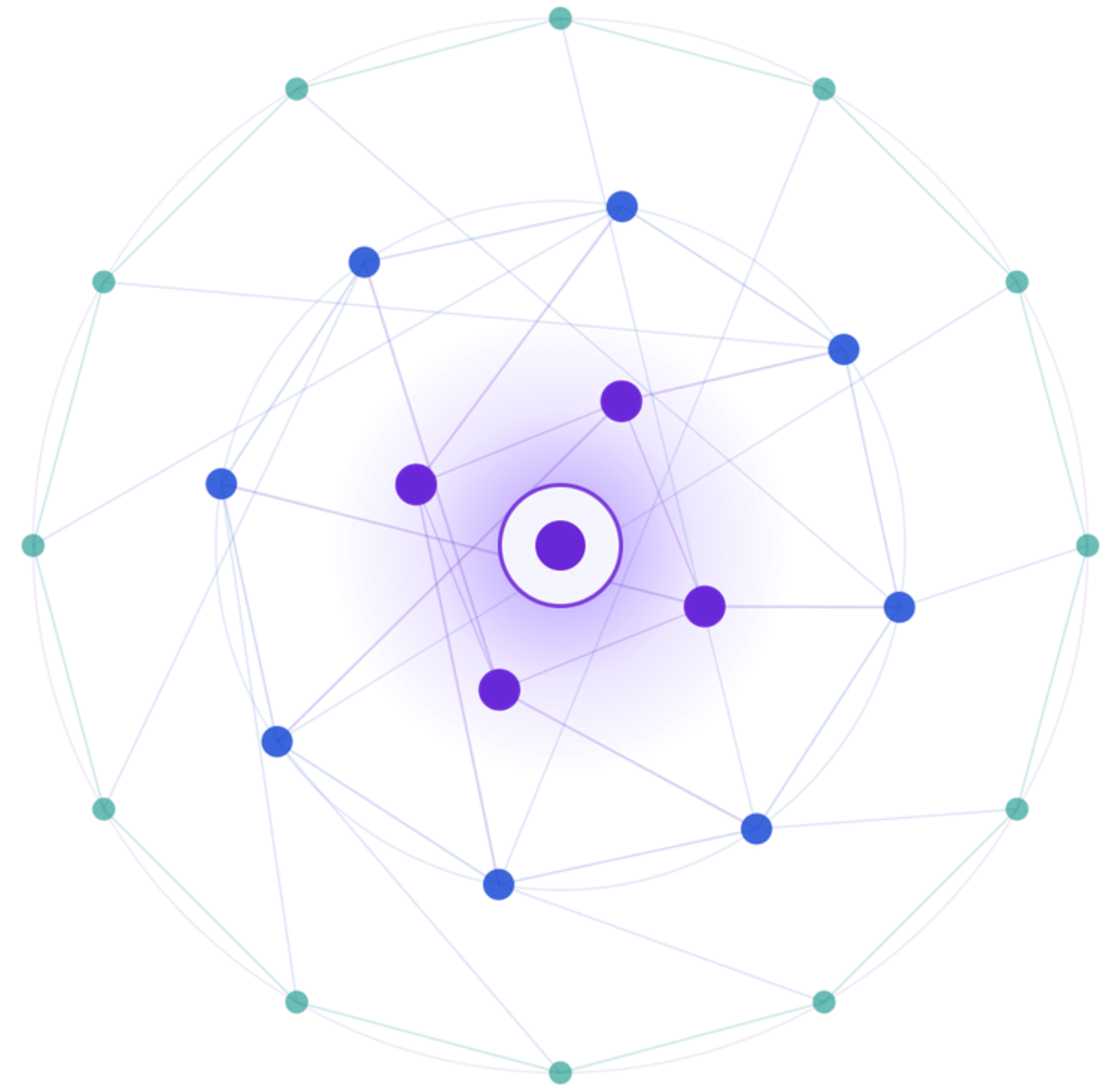


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID -5
- Team Name -Hackhustlers
- College Name -SRMIST,Kattankulathur
- Member Name & Email 1- Aarushi Srivastava [as9551@srmist.edu.in]
- Member Name & Email 2-Samidha Lade [sl2827@srmist.edu.in]
- Member Name & Email 3- Aayati Srivastava[as6909@srmist.edu.in]
- Submission Github link - <https://github.com/samidhalade/duplex-flow>



Theme Real-Time Full-Duplex AI Voice Agents

Theme Focus:

Building intelligent voice agents that can listen, reason, respond, and execute tasks simultaneously for natural, low-latency human–AI interaction.

Key Focus Areas

Full-Duplex Interaction — Simultaneous listening & speaking

Ultra-Low Latency — Real-time conversational response

Multi-Domain AI — Travel, banking, identity & commerce

Tool Orchestration — Execute multi-step tasks through voice

Real-Time Evaluation — Measure latency, accuracy & reliability

Existing Solutions & Gaps

Traditional Voice IVRs / Assistants

- Single-intent command parsing
- Poor handling of multi-step requests
- Limited contextual understanding

Standard LLM Chatbots

- Turn-based interaction
- Users must wait for responses
- Interruptions and corrections feel unnatural

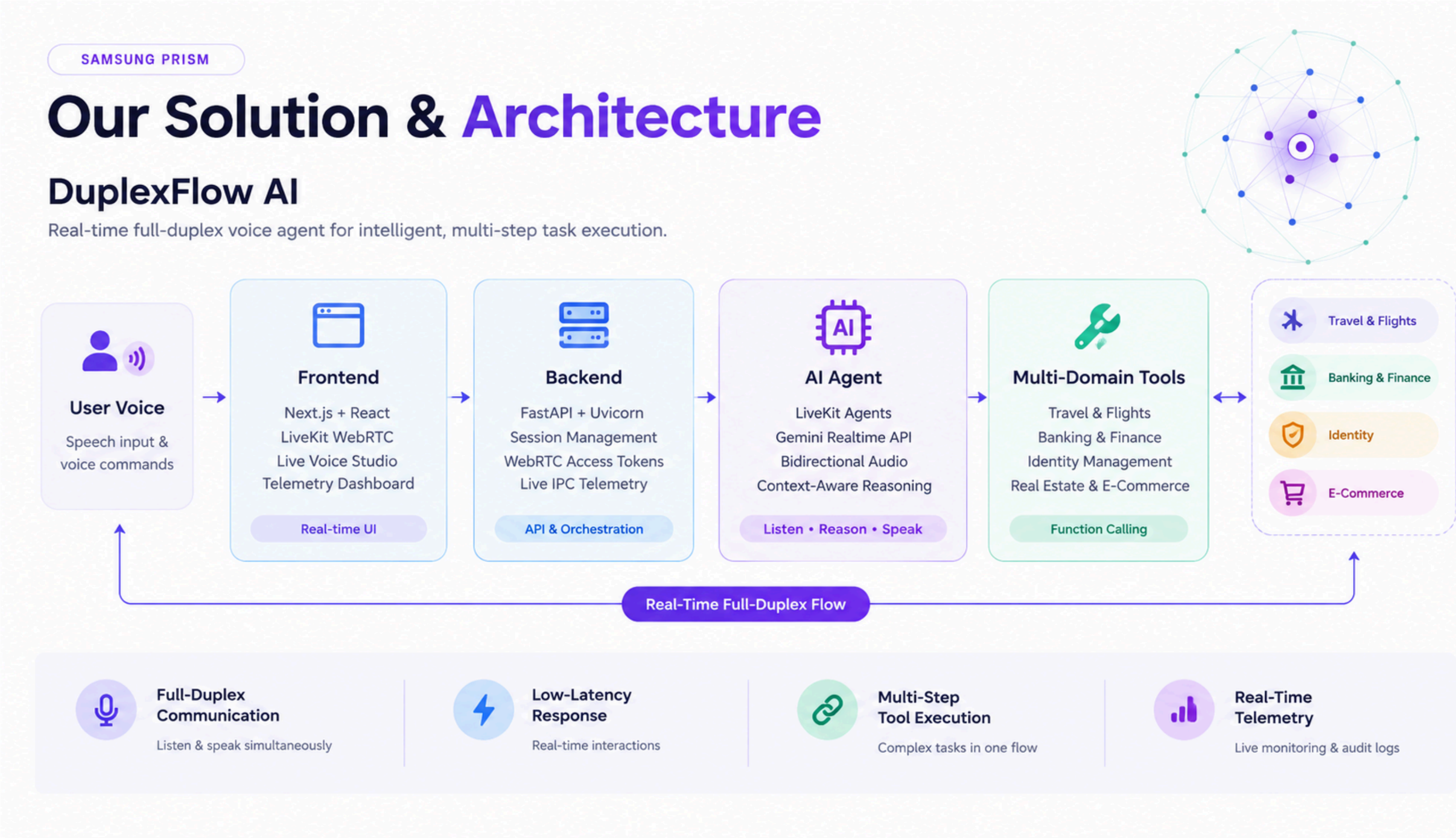
Key Gaps

- No true full-duplex voice interaction
- Poor handling of multi-step commands
- Limited cross-domain task execution
- No real-time tool execution visibility
- Lack of unified latency & accuracy evaluation
- Limited support for context-aware corrections

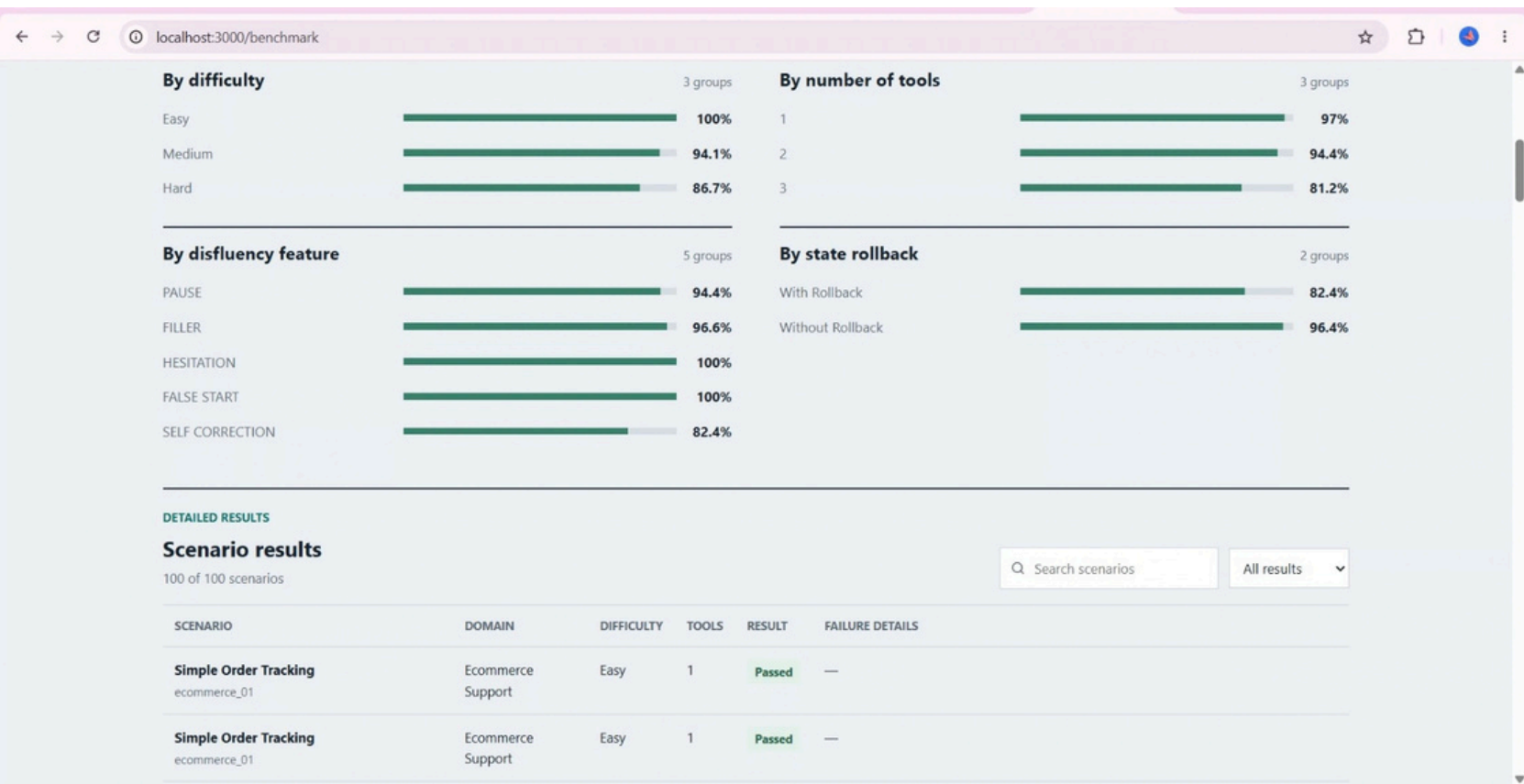
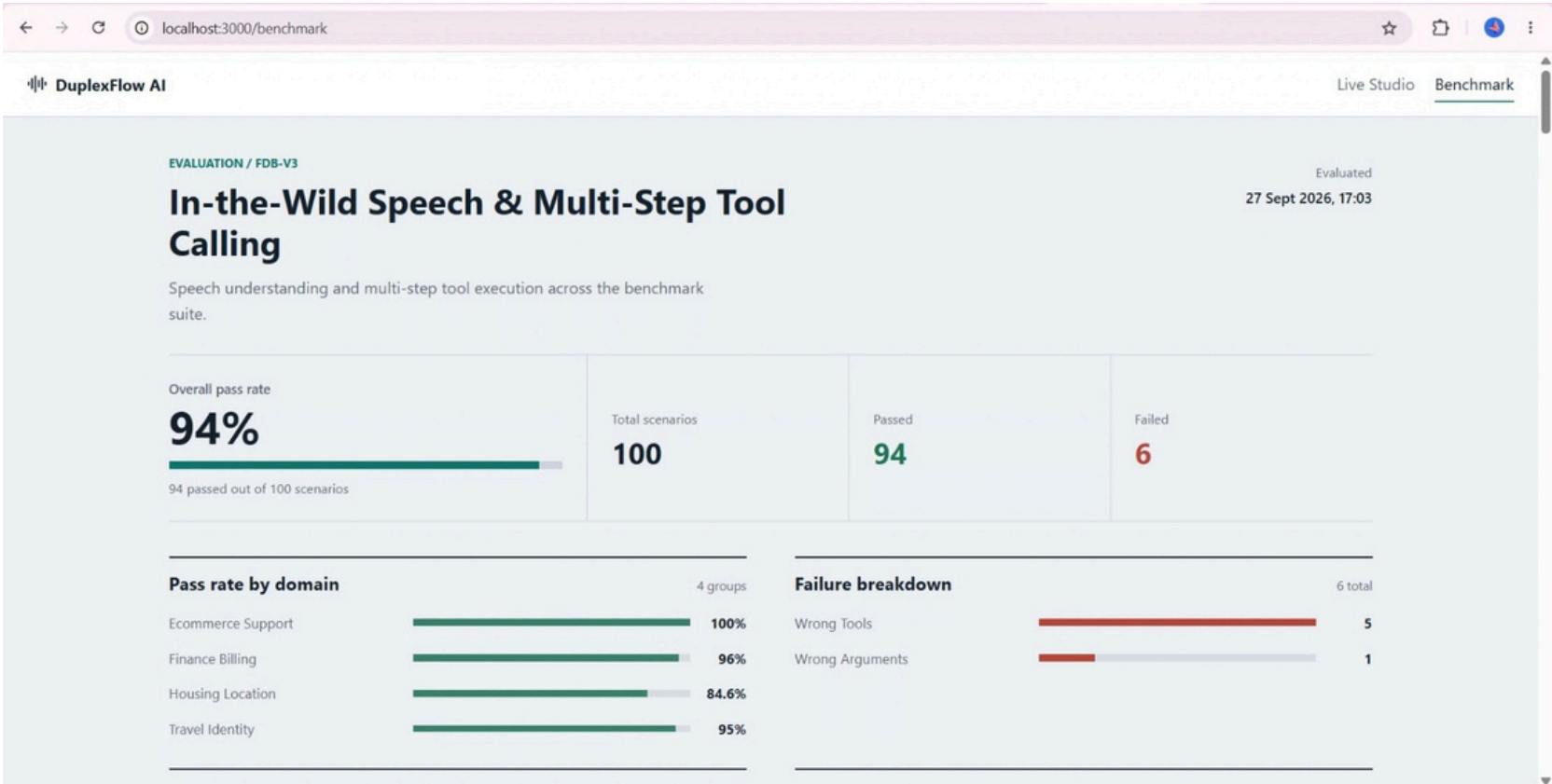
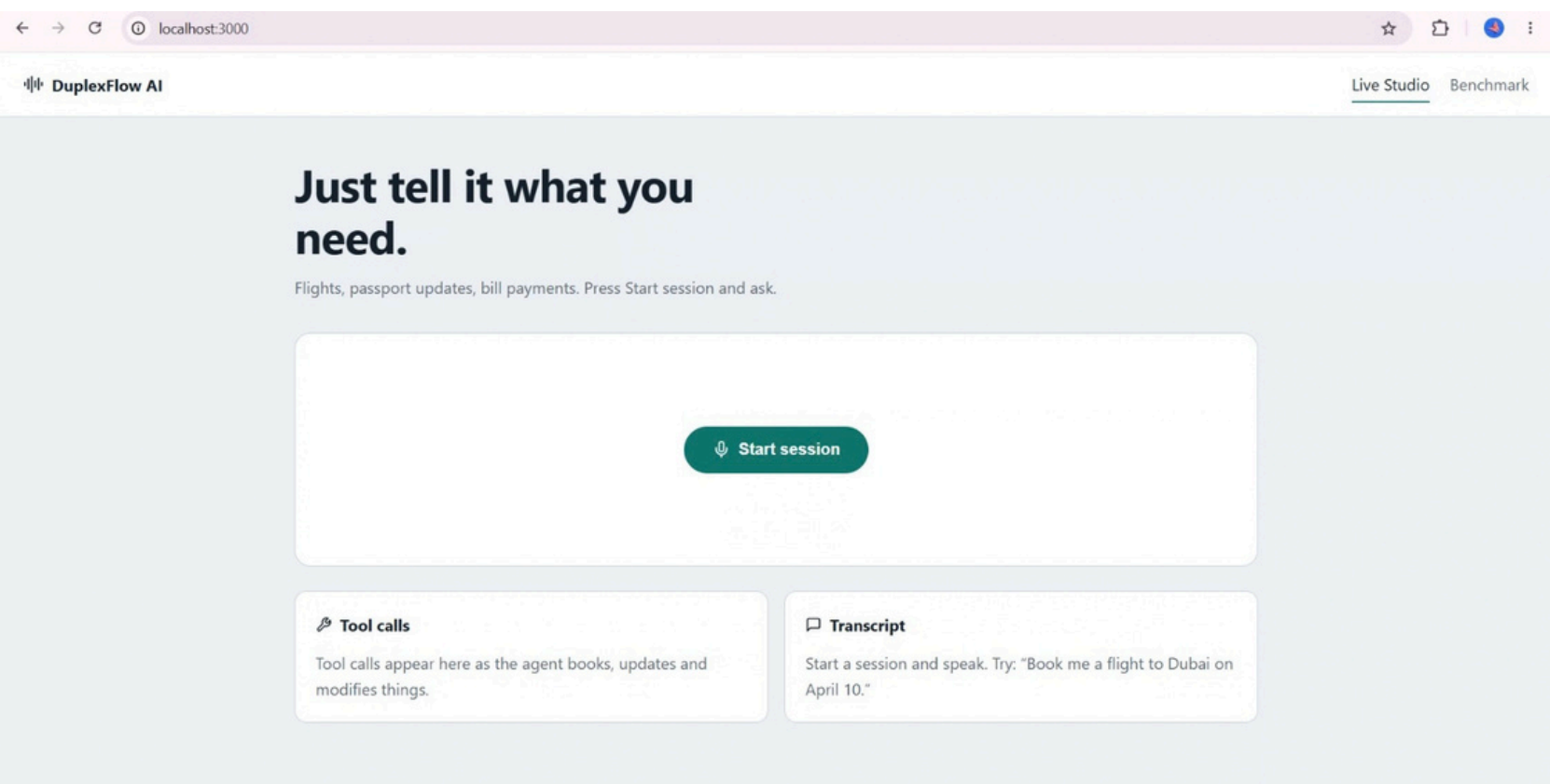
Our Solutions & Architecture Diagram

DuplexFlow AI

A real-time full-duplex voice agent that listens, reasons, speaks, and executes multiple tasks simultaneously.



Demo & Product Walkthrough



The 'Detailed Results' section shows scenario results for 100 of 100 scenarios. The results are categorized by scenario, domain, difficulty, and tools. The pass rates are: Search + Commute Chain (Passed), Double Commute Correction (Passed), Conditional Search Logic (Failed), Multi-Correction Across Statements (Failed), Filler-Heavy Multi-Tool Chain (Passed), Hesitant Three-Tool Chain (Passed), Hesitant Three-Tool Chain (Failed), Full Chain: Filters + Search + Correction (Passed), Simple Flight Search (Passed), Update Passport Number (Passed), and Flight Search with Fillers (Passed).

Scenario	Domain	Difficulty	Tools	Result	Failure Details
Search + Commute Chain	Housing Location	Hard	2	Passed	—
Double Commute Correction	Housing Location	Hard	1	Passed	—
Conditional Search Logic	Housing Location	Hard	3	Failed	Missing tools: ['calculate_commute', 'update_search_filter']
Multi-Correction Across Statements	Housing Location	Hard	2	Failed	Missing tools: ['search_apartments', 'calculate_commute']
Filler-Heavy Multi-Tool Chain	Housing Location	Hard	3	Passed	—
Hesitant Three-Tool Chain	Housing Location	Hard	3	Passed	—
Hesitant Three-Tool Chain	Housing Location	Hard	3	Failed	Missing tools: ['update_search_filter', 'search_apartments', 'calculate_commute']
Full Chain: Filters + Search + Correction	Housing Location	Hard	3	Passed	—
Simple Flight Search	Travel Identity	Easy	1	Passed	—
Update Passport Number	Travel Identity	Easy	1	Passed	—
Flight Search with Fillers	Travel Identity	Easy	1	Passed	—

Tools and tech stack used

AI & Voice: Gemini Realtime ,LiveKit ,WebRTC

Backend: Python, FastAPI,Uvicorn

Frontend: Next.js ,React,Tailwind CSS

Monitoring: Recharts ,Live IPC Telemetry

Real-Time Communication • AI Orchestration • Performance Monitoring

Impact & Use case

Banking & Finance

Voice-based transactions, account services & secure verification.

Travel & Hospitality

Flight search, booking, cancellation & itinerary updates.

E-Commerce & Real Estate

Product/property search, comparison & purchase-related actions.

Identity & Customer Services

Identity verification and multi-step service requests.

Innovation highlights, results and limitations

Innovation

- True Full-Duplex Interaction — Natural interruptions & responses
- Multi-Domain Orchestration — Multiple tasks through voice
- Live Tool Transparency — Real-time visibility of AI actions

Results

- 94% Overall Pass Rate
- 100% Turn-Taking Stability
- 0 Dropped Frames

Limitations

- Background-noise sensitivity
- Requires stable internet/WebRTC connection

What's next

Future Roadmap

- Contextual Memory — Persistent user & task context
- Multimodal AI — Voice + vision capabilities
- Edge Deployment — Optimized for Samsung devices
- Enterprise Scaling — Kubernetes-based deployment

Why DuplexFlow AI Stands Out

- FDB-v3 Evaluation — Dedicated performance benchmarking
- Live IPC Tool Inspector — Real-time visibility into tool execution
- State Rollback — Recovery from failed transactions
- Production-Ready Architecture — Next.js + FastAPI + LiveKit

We don't just build a voice agent — we measure, monitor and control its actions.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo :Yes
- README with reproducible setup instructions :Yes
- Demo video, max 5 minutes : video link(drive).
- Presentation file (PPT or PDF) : Yes

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India